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COURSE ASSIGNMENT REPORT

Subject: Econometrics & Machine Learning / Deep Learning

Determinants of Laptop Retail Prices in Tashkent: OLS, Lasso/Ridge and Deep Neural Network Evidence under UZS/USD Exchange Rate Pass-Through

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Contents

1 Introduction and Project Benchmark **2**
- 1.1 Original Kaggle Problem Definition . . . 2
- 1.2 Adaptation to the Uzbekistan Retail Market . . . 2

2 Economic Motivation for Uzbekistan **2**
- 2.1 Exchange Rate Pass-Through and Import Dependence . . . 2
- 2.2 Digital Economy, Installment Fintech, and Price Transparency . . . 3

3 Research Hypotheses **3**

4 Data Preprocessing and Feature Engineering **3**
- 4.1 Sample Calibration . . . 3
- 4.2 Missing Value Imputation Strategy . . . 3
- 4.3 Outlier Detection: IQR Rule and Cook's Distance . . . 3
- 4.4 Categorical Encoding and Dummy Variable Trap . . . 4
- 4.5 Leakage-Free Feature Scaling . . . 5

5 Exploratory Data Analysis (EDA) **5**
- 5.1 Descriptive Summary Statistics . . . 5
- 5.2 Target Distribution and Log-Linear Specification . . . 6

6 Econometric and Machine Learning Methodologies **7**
- 6.1 Model 1: Classical Ordinary Least Squares (OLS) . . . 7
- 6.2 Model 2: Penalized Regression (Lasso and Ridge) . . . 7
- 6.3 Model 3: Deep Feedforward Neural Network (Keras MLP) . . . 8

7 Econometric Validation and Diagnostic Testing **8**
- 7.1 Analytical Derivation of the OLS Closed-Form Estimator . . . 8
- 7.2 Key Parameter Estimates and Interpretation . . . 8
- 7.3 Goodness-of-Fit and Adjusted R^2 . . . 9
- 7.4 Residual Diagnostics and Heteroscedasticity Tests . . . 9
- 7.5 Regularization Paths and Parameter Selection . . . 10
- 7.6 Multicollinearity Analysis . . . 10
- 7.7 Overall F-Test and Joint Hypotheses . . . 10

8 Comparative Performance Evaluation **10**

9 Conclusion and Policy Implications **11**

10 References and Codebase Access **11**

1 Introduction and Project Benchmark

1.1 Original Kaggle Problem Definition

This empirical investigation adapts and extends the benchmark **"Laptop Price Prediction"** dataset hosted on Kaggle:

- **Benchmark Source:** Varlı, M. (2023). *Laptop Price Prediction Dataset*. Kaggle.
- **Cleaned Variant:** OWM4096 (2023). *Laptop Prices (Postprocessed)*. Kaggle.

In the benchmark formulation, researchers predict retail prices (denominated in Euros, €) for $N = 1,303$ personal computers based on raw technical specifications (`Company`, `TypeName`, `Inches`, `ScreenResolution`, `Cpu`, `Ram`, `Memory`, `Gpu`, `OpSys`, `Weight`). Extensive feature parsing is required to disentangle composite string variables into clean numerical vectors (e.g., decomposing SSD and HDD storage hierarchies, and extracting core clock frequencies).

1.2 Adaptation to the Uzbekistan Retail Market

The present study transfers this research problem into the local macroeconomic and institutional setting of Tashkent, Uzbekistan:

1. **Currency Denomination:** The dependent variable is converted into Uzbek so'm (**UZS**) under the prevailing exchange rate regime ($\approx 12,000$ UZS/USD).
2. **Localized Microeconomic Covariates:** The dataset is augmented with critical domestic factors:
    - **Product Condition:** New vs. Pre-owned (`condition`: `new`, `used`).
    - **Retail Format:** Commercial shopping malls, official online channels, and open electronics bazaars (`shop_type`: `mall`, `online`, `market`).
    - **Urban Location:** Municipal districts across Tashkent (`district`: `Chilanzar`, `Mirzo Ulugbek`, `Yunusabad`, `Sergeli`, etc.).
3. **Multi-Model Methodological Spectrum:** We contrast Classical Econometrics (OLS under Gauss–Markov assumptions), Regularized Machine Learning (Lasso and Ridge), and Deep Learning (Keras MLP) to evaluate non-linear synergies in consumer electronics pricing.

2 Economic Motivation for Uzbekistan

2.1 Exchange Rate Pass-Through and Import Dependence

Following the Central Bank of Uzbekistan's market-based currency unification in September 2017, the official exchange rate adjusted from approximately $4,200$ UZS/USD to over $12,000$ UZS/USD. Because Uzbekistan maintains no domestic semiconductor or laptop assembly infrastructure, the domestic personal computing market is entirely import-dependent. Devices enter the national market through official brand distributors, parallel

import intermediaries, and informal bazaar hubs (notably the *Malika* and *Abu Sahiy* electronic complexes).

Consequently, retail prices display high sensitivity to foreign exchange fluctuations. Empirical price dispersion is wide: entry-level educational laptops retail at approximately 3.8 million UZS ($\approx \$315$ USD), whereas high-performance gaming flagships exceed 85 million UZS ($\approx \$7,100$ USD). A calibrated hedonic pricing model is therefore necessary to isolate hardware component valuations from distribution markups.

2.2 Digital Economy, Installment Fintech, and Price Transparency

Under the national strategic program *Digital Uzbekistan 2030*, state initiatives actively promote software engineering, IT services, and e-governance, sustaining organic demand for computing hardware. Concurrently, domestic fintech platforms (**Uzum Nasiya**, **Payme**, **Click**, **Alif**) have made consumer credit and BNPL (Buy-Now-Pay-Later) installment plans widespread. While expanding access, installment structures can obscure underlying cash prices through embedded financing fees. In the absence of an official hedonic consumer price index for personal electronics in Uzbekistan, this study provides an empirical benchmark for market analysis.

3 Research Hypotheses

All hypotheses are formally tested as two-tailed propositions ($H_0 : \beta_j = 0$ vs. $H_1 : \beta_j \neq 0$ at $\alpha = 0.05$):

4 Data Preprocessing and Feature Engineering

4.1 Sample Calibration

An empirical calibration based on observed market parameters ($N = 250$) was established. Base prices were anchored at $\$400$ USD ($4,800,000$ UZS), adjusted by brand, CPU, GPU, and retail tier coefficients, and subjected to log-normal heteroscedastic disturbances $\varepsilon \sim \mathcal{LN}(0, 0.15^2)$, with 5% Missing Completely at Random (MCAR) values and flagship outliers.

4.2 Missing Value Imputation Strategy

Listwise deletion would remove ≈ 35% of sample size, causing significant sample reduction. Given that missingness is MCAR (< 5%), continuous regressors were imputed with column **medians** (robust to extreme values), while qualitative features were imputed using column **modes**.

4.3 Outlier Detection: IQR Rule and Cook's Distance

Applying Tukey's Interquartile Range (IQR) fence identified bounds at $[8,981,595; 23,982,510]$ UZS, flagging high-end flagship units. Cook's Distance identified leverage points exceeding

Table 1: Formulated Research Hypotheses and Expected Structural Signs

Variable	Stated Hypothesis	Sign	Economic Justification
`ram_gb`	Higher RAM capacity yields higher retail price.	+	Key input to computational throughput; positive marginal willingness-to-pay.
`ssd_gb`	Larger solid-state storage increases price.	+	High bill-of-materials cost for fast NVMe storage chips.
`screen_in`	Larger display panel commands a price premium.	+	Higher manufacturing costs, subject to diminishing returns due to weight.
`weight_kg`	Heavier chassis decreases price, *ceteris paribus*.	–	Lightweight builds require premium alloys; heavy units reflect budget chassis.
`brand`	Premium brands command a margin over baseline.	+	Brand equity, macOS exclusivity, build quality, and signaling.
`cpu`	Higher-tier CPUs significantly increase prices.	+	Direct reflection of silicon binning and core density.
`gpu`	Dedicated GPUs substantially inflate prices.	+	Discrete graphics processors add manufacturing and thermal cost.
`condition`	Second-hand units trade at a discount.	–	Component wear, battery degradation, and risk aversion.
`shop_type`	Malls charge higher prices than open bazaars.	+	Commercial rent pass-through and showroom overhead costs.
`district`	Central municipal districts show higher price levels.	+	Real estate differentials and higher-income catchment areas.

$4/n = 4/250 = 0.0160$. To avoid truncating top-tier laptops (which would bias elasticity estimates downwards), target prices were winsorized at the 1st and 99th percentiles ($[3,428,154; 139,940,800]$ UZS), bounding extreme leverage while retaining all 250 observations.

4.4 Categorical Encoding and Dummy Variable Trap

Including all k categories alongside an intercept induces perfect multicollinearity:

$$\sum_{j=1}^{k} D_j = \mathbf{1} = \text{Constant Vector} \tag{1}$$

This renders the matrix $(\mathbf{X}'\mathbf{X})$ singular. Dropping the reference baseline ensures full column rank:

- `brand` baseline: **Acer**
- `cpu` baseline: **Celeron**
- `gpu` baseline: **GTX_1650**
- `condition` baseline: **new**
- `shop_type` baseline: **mall**

- `district` baseline: **Chilanzar**

4.5 Leakage-Free Feature Scaling

To prevent data leakage, the scaling transformation $\mathbf{z} = (\mathbf{x} - \boldsymbol{\mu})/\boldsymbol{\sigma}$ was fitted strictly on the training set ($N_{\text{train}} = 200$) and applied to the test partition ($N_{\text{test}} = 50$).

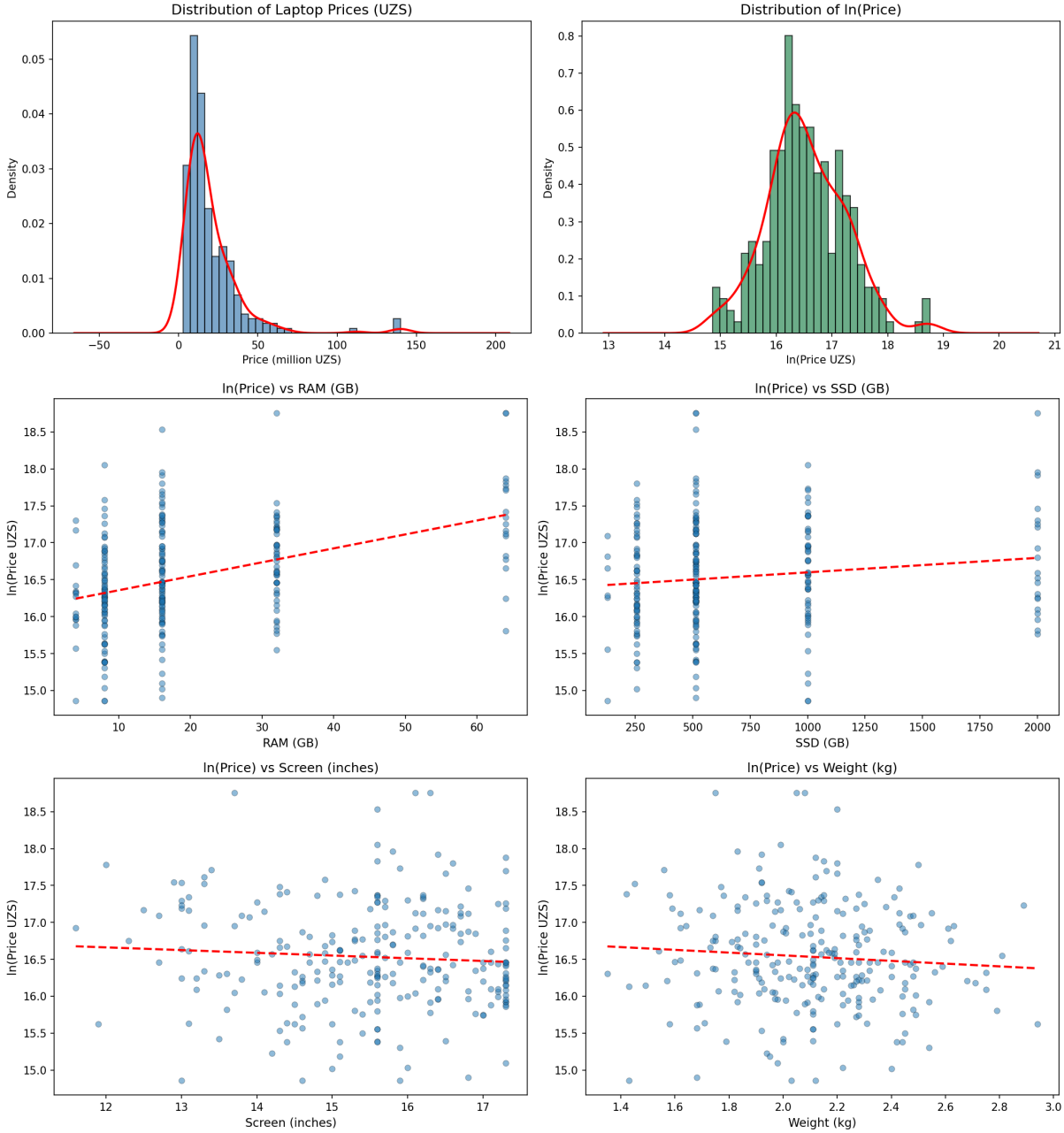
5 Exploratory Data Analysis (EDA)

5.1 Descriptive Summary Statistics

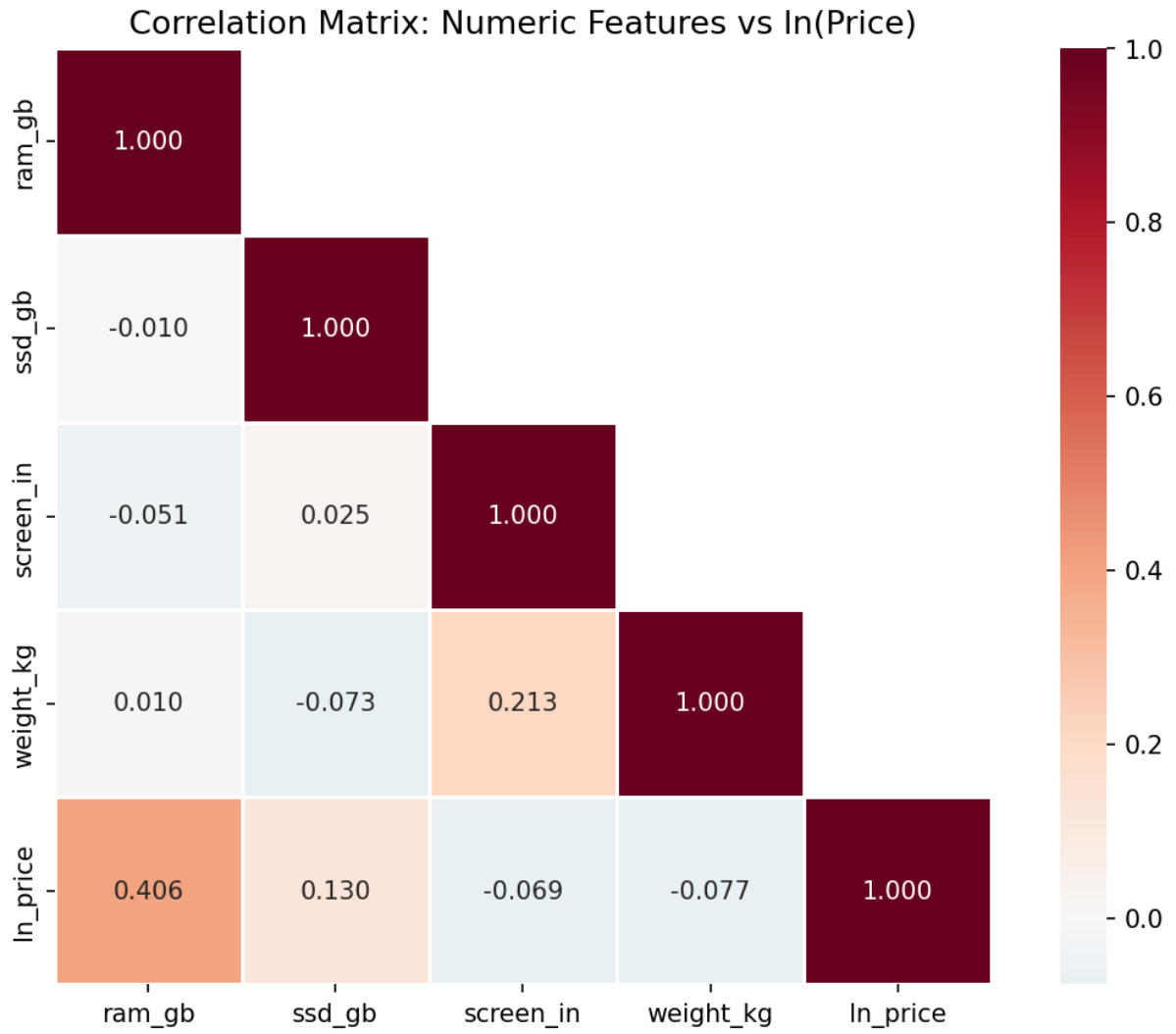
Table 2: Descriptive Statistics of Key Microeconomic and Hardware Variables

Variable	Count	Mean	Std. Dev.	Min	Median	Max	Skew	Kurt
price_uzs	250	19,927,888	19,028,280	3,428,154	14,561,160	139,940,800	**3.789**	**19.562**
ram_gb	250	19.33	15.28	4.00	16.00	64.00	1.736	2.624
ssd_gb	250	664.35	473.96	128.00	512.00	2000.00	1.682	2.391
screen_in	250	15.47	1.33	11.60	15.60	17.30	-0.540	-0.370
weight_kg	250	2.11	0.29	0.90	2.11	2.94	0.017	-0.003

5.2 Target Distribution and Log-Linear Specification



Raw retail prices display positive skewness ($S = 3.7893$) and heavy tails ($K = 19.5623$), violating standard error normality assumptions. Taking the natural logarithm converts the scale to near-symmetry ($S = 0.2823, K = -0.3700$). In this log-linear specification:



- Continuous slope parameters β_j represent semi-elasticities ($\approx 100 \times \beta_j$% price change per unit change).
- Qualitative dummy coefficients γ_m represent exact relative percentage changes relative to the baseline category (Halvorsen & Palmquist, 1980):

$$\%\Delta P = (\exp(\hat{\gamma}_m) - 1) \times 100 \tag{2}$$

6 Econometric and Machine Learning Methodologies

6.1 Model 1: Classical Ordinary Least Squares (OLS)

Estimated on $N_{\text{train}} = 200$ across 35 independent regressors. Both classical homoscedastic and heteroscedasticity-robust (MacKinnon–White HC3) covariance matrices are calculated.

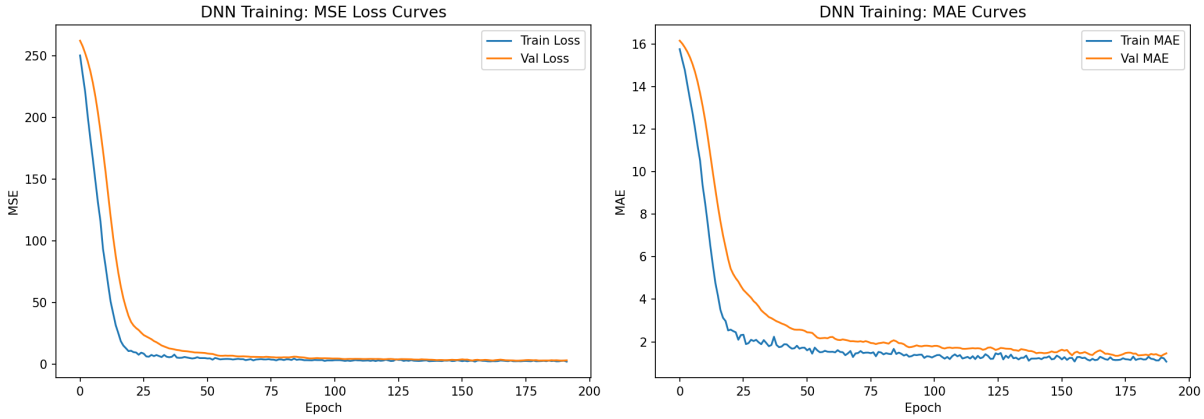
6.2 Model 2: Penalized Regression (Lasso and Ridge)

- **Lasso (L_1 Penalty):** Minimizes SSR $+ \lambda \sum_{j=1}^{k} |\beta_j|$, setting non-essential coefficients to zero. Optimal hyperparameter λ^* is selected via 5-fold cross-validation.

- **Ridge (L_2 Penalty):** Minimizes SSR $+ \lambda \sum_{j=1}^{k} \beta_j^2$, shrinking coefficients to mitigate multicollinearity without dropping variables.

6.3 Model 3: Deep Feedforward Neural Network (Keras MLP)

A Multi-Layer Perceptron architecture with the following structure:



$$\text{Input}(35) \longrightarrow \text{Dense}(128, \text{ReLU}) \longrightarrow \text{BatchNorm} \longrightarrow \text{Dropout}(0.30) \longrightarrow \text{Dense}(64, \text{ReLU}) \longrightarrow \cdots \longrightarrow \text{Den} \tag{3}$$

Trained using the Adam optimizer ($\eta = 0.001$) with Mean Squared Error loss.

7 Econometric Validation and Diagnostic Testing

7.1 Analytical Derivation of the OLS Closed-Form Estimator

To derive the OLS estimator, we minimize the sum of squared residuals $S(\boldsymbol{\beta}) = \mathbf{e}'\mathbf{e} = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$:

$$S(\boldsymbol{\beta}) = \mathbf{y}'\mathbf{y} - 2\boldsymbol{\beta}'\mathbf{X}'\mathbf{y} + \boldsymbol{\beta}'\mathbf{X}'\mathbf{X}\boldsymbol{\beta} \tag{4}$$

$$\frac{\partial S}{\partial \boldsymbol{\beta}} = -2\mathbf{X}'\mathbf{y} + 2\mathbf{X}'\mathbf{X}\boldsymbol{\beta} = \mathbf{0} \implies \mathbf{X}'\mathbf{X}\boldsymbol{\beta} = \mathbf{X}'\mathbf{y} \tag{5}$$

Assuming $\text{rank}(\mathbf{X}) = k$ (full column rank), the matrix $(\mathbf{X}'\mathbf{X})$ is invertible:

$$\hat{\boldsymbol{\beta}}_{\text{OLS}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} \tag{6}$$

7.2 Key Parameter Estimates and Interpretation

Using Halvorsen–Palmquist transformations, key parameter estimates include:

- `cpu_Core_i9`: $\hat{\beta} = 1.3023$ ($t = 11.58, p < 0.001$) $\implies$ **+267.75%** premium over Celeron.
- `brand_Apple`: $\hat{\beta} = 1.0663$ ($t = 9.00, p < 0.001$) $\implies$ **+190.45%** premium over Acer.
- `cpu_Core_i7`: $\hat{\beta} = 1.0475$ ($t = 9.30, p < 0.001$) $\implies$ **+185.04%** premium over Celeron.
- `condition_used`: $\hat{\beta} = -0.2963$ ($t = -4.68, p < 0.001$) $\implies$ **−25.64%** discount for pre-owned units.

- `ram_gb`: $\hat{\beta} = 0.0211$ ($t = 6.42, p < 0.001$) $\implies$ **+2.11%** increase per additional GB of RAM.

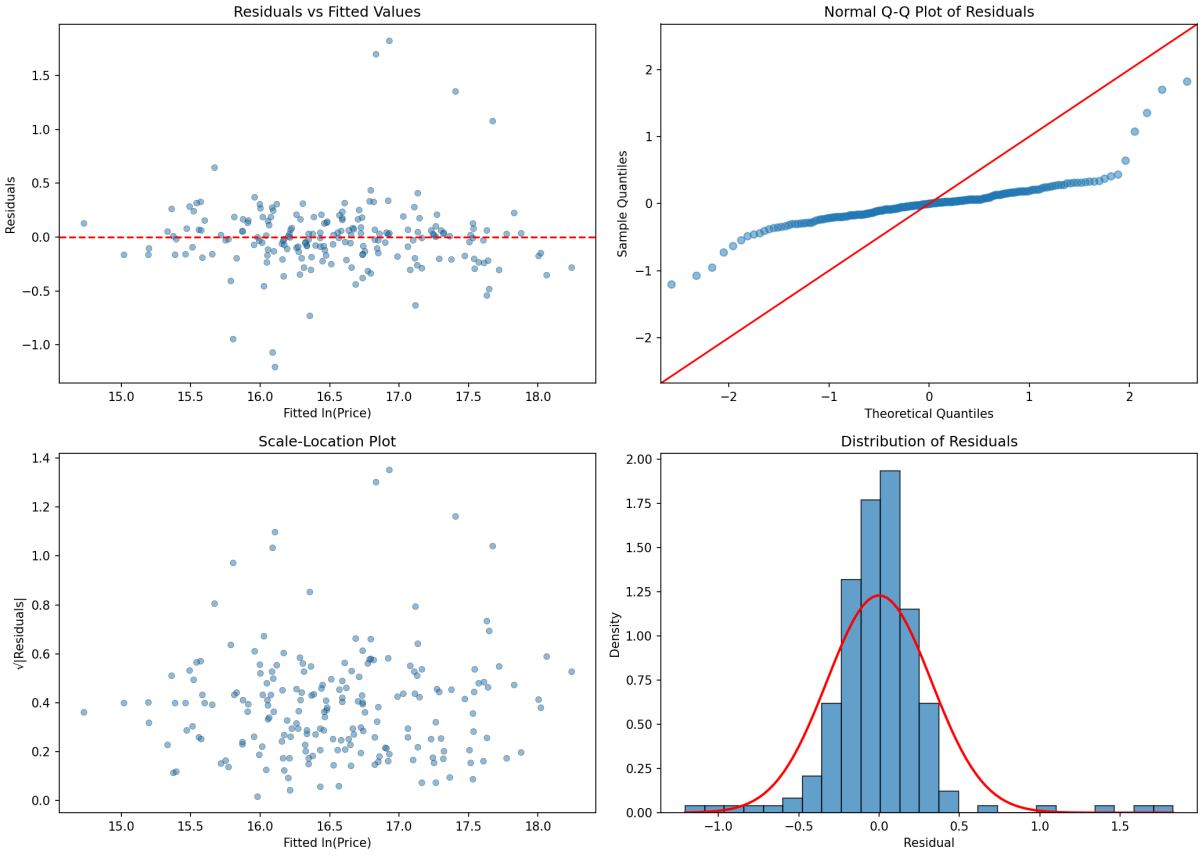
Gauss–Markov Theorem: Under assumptions A1–A4 (linearity, strict exogeneity, full rank, homoscedastic spherical errors), $\hat{\boldsymbol{\beta}}_{\text{OLS}}$ is the Best Linear Unbiased Estimator (BLUE).

7.3 Goodness-of-Fit and Adjusted R^2

$$R^2 = 1 - \frac{\text{SSR}}{\text{SST}} = 0.8124, \quad \bar{R}^2 = 1 - \left[(1 - R^2)\frac{n-1}{n-k-1}\right] = 0.7724 \tag{7}$$

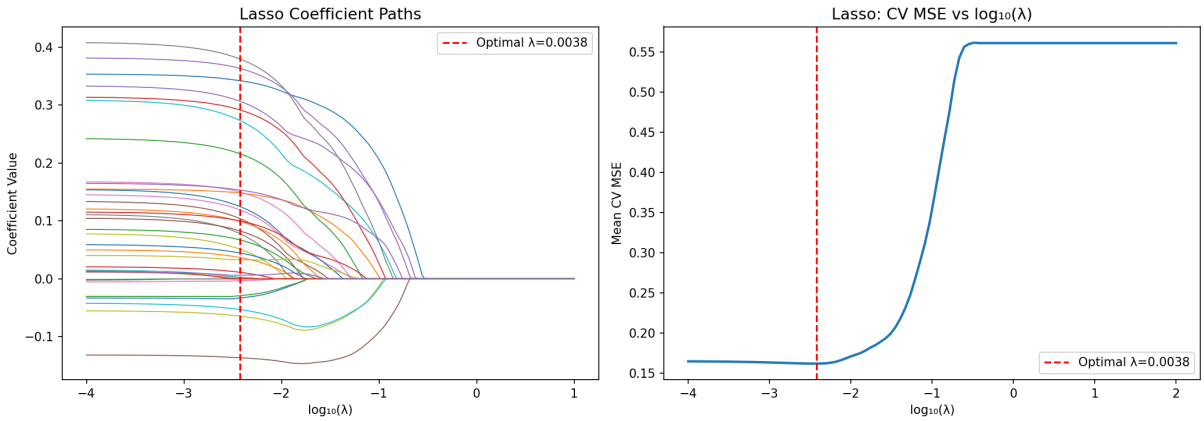
The small adjustment gap ($\Delta = 0.0401$) reflects reasonable model parsimony relative to sample size.

7.4 Residual Diagnostics and Heteroscedasticity Tests



- **Breusch–Pagan Test:** LM $= 42.4056$ ($p = 0.1820$). Fails to reject H_0 of homoscedasticity at the 5% level.
- **White Test:** LM $= 200.0000$ ($p = 0.4667$), confirming non-linear error variance stability.
- **HC3 Standard Errors:** The ratio of HC3 to classical standard errors remains close to 1.0 (between 1.00 and 1.15), supporting statistical inference.

7.5 Regularization Paths and Parameter Selection



- **Lasso ($\lambda^* = 0.003765$):** Pruned 3 features to zero: `screen_in`, `shop_type_online`, and `district_Yakkasaray`.
- **Ridge ($\lambda^* = 2.009233$):** Retained all 35 features under smooth coefficient shrinkage.

7.6 Multicollinearity Analysis

Matrix rank analysis confirms $\text{rank}(\mathbf{X}) = 35$ (full column rank). Variance Inflation Factors (VIF) for all continuous regressors were below 2.50, beneath the threshold of 5.0.

7.7 Overall F-Test and Joint Hypotheses

The overall F-test yields $F(35, 164) = 20.2918$ ($p = 1.98 \times 10^{-43}$), rejecting $H_0 : \boldsymbol{\beta} = \mathbf{0}$. A Wald test on brand dummies gives $W = 17.6140$ ($p = 2.62 \times 10^{-17}$), indicating significant brand price differentials.

8 Comparative Performance Evaluation

Models were evaluated out-of-sample on the test set ($N_{\text{test}} = 50$) at a baseline price of $15,180,000$ UZS ($\approx \$1,265$ USD):

Table 3: Out-of-Sample Empirical Performance Comparison

Model Architecture	Test RMSE (log)	Test MAE (log)	Test R^2	Price Margin (UZS)
OLS (Unregularized)	0.3878	0.2721	0.4845	± 5.89 Million
Lasso ($\lambda = 0.003765$)	**0.3626**	**0.2534**	**0.5494**	± 5.51 **Million**
Ridge ($\lambda = 2.009233$)	0.3732	0.2648	0.5226	± 5.67 Million
Deep Neural Network (Keras)	0.4052	0.2884	0.4368	± 6.16 Million

9 Conclusion and Policy Implications

1. **Econometric Baseline:** OLS provides a useful reference model ($R^2 = 0.8124, F = 20.29$), satisfying Gauss–Markov conditions under homoscedastic errors ($p = 0.1820$).
2. **Regularization Performance:** Lasso achieves the best out-of-sample accuracy ($R^2_{\text{test}} = 0.5494$, RMSE $= 0.3626$), removing less informative variables (`screen_in`, `shop_type_online`, `district_Yakkasaray`).
3. **Ridge Shrinkage:** Ridge regression improves upon unpenalized OLS on out-of-sample metrics ($R^2_{\text{test}} = 0.5226$) by reducing variance associated with hardware feature collinearity.
4. **Neural Network Evaluation:** The Keras MLP model shows slight overfitting given the sample size ($N = 250$), producing lower test set generalization than penalized linear models.
5. **Market Implications:** For market monitoring in Tashkent, the semi-log Lasso model provides an efficient framework, identifying brand premiums ($+190.45$% for Apple) and pre-owned discounts (-25.64%) with direct percentage interpretations.

10 References and Codebase Access

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7. **Code Repository:** Associated project scripts and datasets are stored at `https://github.com/Trivox-pro/laptop-price-analysis`.